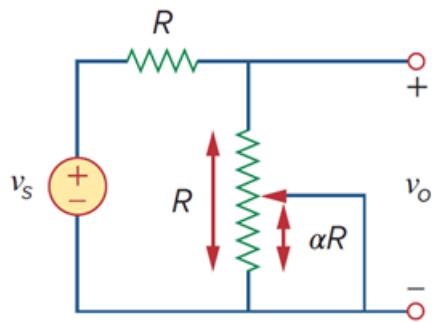


Problem

In the circuit in Fig. 2.137, the wiper divides the potentiometer resistance between αR and $(1 - \alpha)R$, $0 \leq \alpha \leq 1$. Find v_o/v_s .

Figure 2.137:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

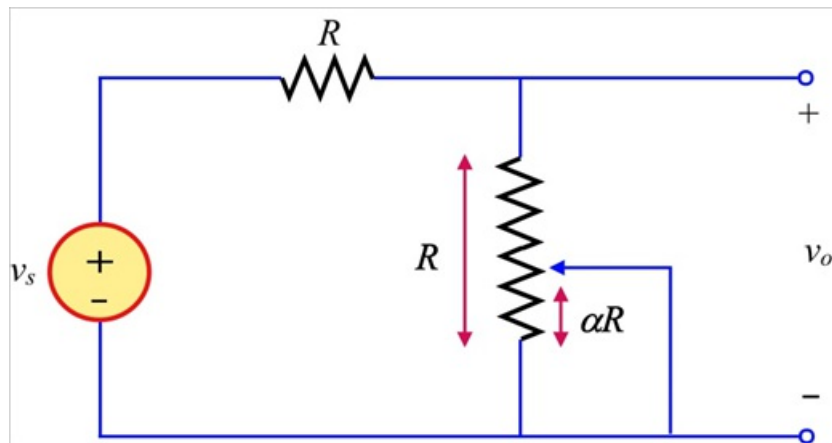


Figure 1

Step 2 of 5

Consider the wiper divides resistance between αR and $(1-\alpha)R$.

The value of α varies $0 \leq \alpha \leq 1$.

The modified diagram of Figure 1 is shown in Figure 2:

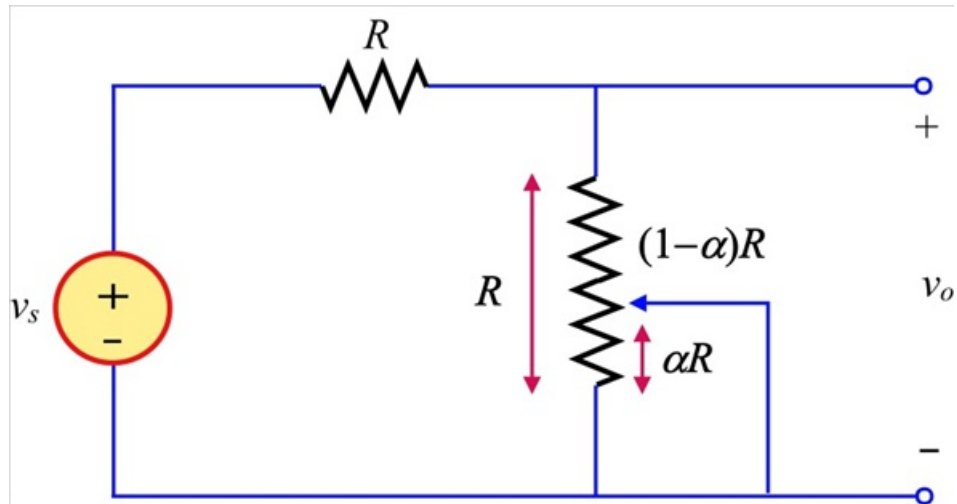


Figure 2

Step 3 of 5

From Figure 2, the potential divider is connected to the $(1-\alpha)R$.

The modified diagram of Figure 2 is shown in Figure 3:

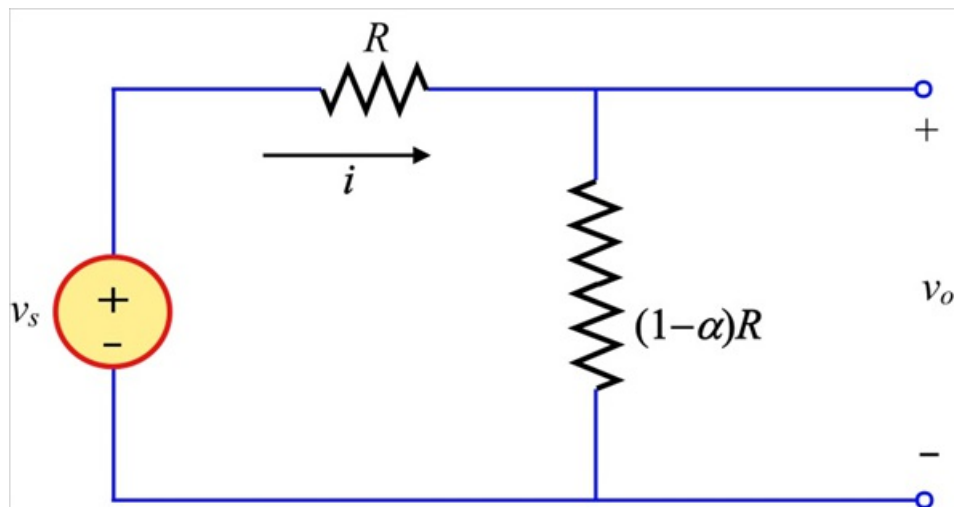


Figure 3

Step 4 of 5

From Figure 3, apply Kirchhoff's voltage law.

$$-v_s + Ri + i(1-\alpha)R = 0$$

$$v_s = Ri + i(1-\alpha)R$$

$$= (R + R - \alpha R)i$$

$$= (2R - \alpha R)i$$

$$v_s = (2 - \alpha)Ri \dots\dots (1)$$

Step 5 of 5

Calculate the output voltage as follows.

$$v_o = (1 - \alpha) Ri \dots\dots (2)$$

Divide equation (2) with equation (1).

$$\begin{aligned} \frac{v_o}{v_s} &= \frac{(1 - \alpha) Ri}{(2 - \alpha) Ri} \\ &= \frac{1 - \alpha}{2 - \alpha} \end{aligned}$$

Therefore, the expression for $\frac{v_o}{v_s}$ is $\boxed{\frac{1 - \alpha}{2 - \alpha}}$.